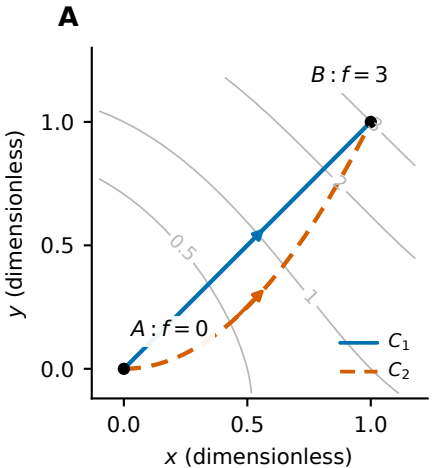
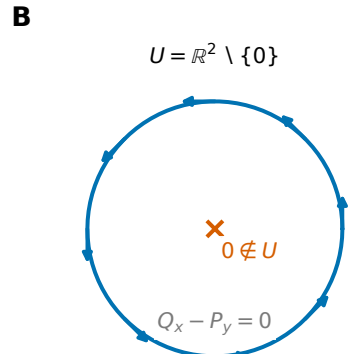


$$\int_{C_1} \nabla f \cdot \mathrm{d}\mathbf{r} = \int_{C_2} \nabla f \cdot \mathrm{d}\mathbf{r} = 3$$



$$\oint_C \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2} = 2\pi \neq 0$$